Temporal Avian Contact Networks for Epidemic Simulation and Prediction: Centrality and Community Structures

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In this study, we enhance traditional centrality metrics to account for temporal dynamics in avian contact networks, introducing new measures like Time-Weighted Degree Centrality with Decay, and improved community detection algorithms. Our findings, validated through infection simulations, demonstrate the critical role of these temporal metrics in accurately predicting disease spread and informing effective wildlife epidemic control strategies.

Community Detection | Centrality | Network Epidemiology | Simulation

Significance Statement

This study investigates temporal centrality measures, designs new metrics, and explores community detection algorithms on the AWN dataset. By running infection simulations, we demonstrate the significance of these measures in predicting disease spread. Our novel metrics, including Time-Weighted Degree Centrality with Decay, provide a more accurate depiction of nodes' importance in dynamic networks. We also explore the inference of these concepts, such as the typical positions of high-centrality nodes within communities. This research advances theoretical network analysis and offers practical implications for designing effective disease control strategies in wildlife populations, highlighting the limitations of static measures and the benefits of considering temporal dynamics.

Understanding the dynamics of disease spread within avian populations is crucial for effective epidemic control, as birds play a significant role in the transmission of various pathogens. Traditional centrality metrics often fail to account for the temporal nature of interactions in dynamic networks, leading to inaccurate assessments of nodes' importance. In this study, we analyze the Aves Wildbirds Network dataset to address this gap by investigating temporal centrality measures and designing new metrics that better capture the dynamic importance of nodes. We introduce the Time-Weighted Degree Centrality (TWDC) and its recursive variant with decay, which incorporate both the weight and timing of interactions. By applying improved community detection techniques, we uncover structural insights and examine the typical positions of high-centrality nodes within communities. Our infection simulations validate the effectiveness of these centrality measures in predicting disease spread. The results highlight the limitations of static centrality metrics and underscore the importance of temporal dynamics in network analysis. This research provides valuable theoretical advancements and practical tools for designing targeted disease control strategies in wildlife populations, ultimately contributing to better management of avian-borne diseases.

Methods and Materials

Data Preparation. The original dataset used in this study is the Aves Wildbird Network (AWN) dataset (1), which represents interactions among wild birds observed at specific locations over six consecutive days. The network data is in the form of edge list with weights and labels (day of observation) in the third and the forth column respectively.

To facilitate a disease spread simulation over a longer time period in the wildbird population context, we generated synthetic data for a duration of 100 days while preserving the network properties of the original 6 days. Specifically, for each day, given the number of nodes (birds) and edges (interactions) per day to match the original data distribution, we randomly connected pairs of nodes, simulating new interactions between birds on each day. The connections are assigned random weights between 0.01 and 0.15.

Community Detection. Community detection methods are able to locate communities in a complex network structure. The identified communities are groups of nodes closely connected by internal edges representing certain types of relationships. It is widely adopted in network analysis for both single layer model and multiplex structure like social network, in which connections are stratified into layers indicating different types of relationships.

To explore the structural information of the AWN dataset, we applied both single-layer and multi-layer community detection.

Single Layer Detection.

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Louvain algorithm. The Louvain algorithm has two main steps. In the first phase of the Louvain algorithm, known as modularity optimization, each node is initially assigned to its own community (2). For each node i, we calculate the change in modularity ΔQ by removing i from its community and placing it in the community of each neighbor j. Then, we place the node i in the community that results in the greatest increase in modularity, but only if ΔQ is positive. If no positive possible gain, i remains in the original community. We apply this process repeatedly and sequentially for all nodes until we cannot achieve any further improvement (3). The process of calculation on changes in modularity ΔQ when we move node i into the community of node j as:

$$\Delta Q = \left[\frac{\Sigma_{in} + k_{i,in}}{2m} - \left(\frac{\Sigma_{tot} + k_i}{2m}\right)^2\right] - \left[\frac{\Sigma_{in}}{2m} - \left(\frac{\Sigma_{tot}}{2m}\right)^2 - \left(\frac{k_i}{2m}\right)^2\right] \quad [1]$$

where Σ_{in} is the sum of the weights of the edges inside the community C, Σ_{tot} is the sum of the weights of the edges incident to nodes in C, k_i is the sum of the weights of the edges incident to node i, $k_{i,in}$ is the sum of the weights of the edges from i to nodes in C, and m is the sum of the weights of all the edges in the network.

In the second phase, known as community aggregation, a new network is built whose nodes are the communities found in the first phase. The weights of the edges between the new nodes are the sum of the weights of the edges between nodes in the corresponding communities. Once this phase is complete, we reapply the first phase of the algorithm to the new network, and the process is repeated iteratively until there are no more changes in the community structure (4).

Leiden Algorithm. The Leiden algorithm is a refinement of the Louvain method, addressing some of its limitations and enhancing the quality of community detection. This algorithm is structured into three main phases, designed to improve both the accuracy and resolution of identified communities (5). Since the first and third phases are similar to the Louvain method, we will focus on the distinctive middle phase of the Leiden algorithm which differentiates it significantly from its predecessor.

Refinement Phase. Given a locally optimized partition from first phase, the Leiden algorithm applies a unique refinement phase. In this phase, each detected community is reevaluated for potential internal structure by applying the local moving method within the community itself. This allows the algorithm to split communities into smaller, more cohesive sub-communities. Then, the algorithm enhances the overall resolution and reveals finer-grained community structures that the Louvain method may overlook.

ModularityVertexPartition. This method focuses on optimizing the traditional modularity measure, Q, which quantifies the density of links within communities compared to links between communities. The formula for Q is given by:

$$Q = \frac{1}{2m}\sum_{ij}\left[A_{ij} - \frac{k_i k_j}{2m}\right]\delta(c_i, c_j) \quad [2]$$

where A_{ij} is the weight of the edge between nodes i and j, k_i and k_j are the sums of the weights of the edges attached to nodes i and j, respectively, m represents the total weight of all edges in the network, and $\delta(c_i, c_j)$ is the Kronecker delta function that outputs 1 if nodes i and j are in the same community, and 0 otherwise.

A higher modularity indicates a stronger division of the network into communities. The ModularityVertexPartition computes Q by contrasting the actual number of intracommunity edges against the expected number in a randomly connected network, based on node degrees. This approach seeks to maximize Q, promoting the formation of densely interconnected communities with fewer connections between them.

RBConfigurationVertexPartition. The RBConfigurationVertexPartition adds a resolution parameter to the modularity optimization, making it particularly effective in networks with varied node degree distributions shown as:

$$Q = \frac{1}{2m}\sum_{ij}\left[A_{ij} - \gamma\frac{k_i k_j}{2m}\right]\delta(c_i, c_j) \quad [3]$$

This method adapts the standard modularity formula by incorporating a resolution parameter that adjusts the sensitivity of the algorithm to community scale. It normalizes the influence of each node's degree on modularity, ensuring that nodes with high degrees do not overshadow the community structure. This adjustment allows the detection of communities at multiple scales, broadening the Leiden algorithm's applicability across different network sizes and types.

Multilayer Detection. To look into more potential implications of the applications of community detection algorithms on AWN dataset, we analyzed the interpretability of the algorithm known as Weighted Simultaneous Symmetric Non-Negative Matrix Tri-Factorization (WSSNMTF). The formulation of the problem is based on the Non-Negative Matrix Tri-Factorization (NMTF) framework in its symmetric form:

$$\mathbf{A}^{(i)} \approx \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T, \quad \text{where } i \in \{1, \ldots, N\} \quad [4]$$

Say we have n nodes and k communities. Then, $\mathbf{A}^{(i)} \in \mathbb{R}_+^{n\times n}$ is the symmetric adjacency matrix for layer i that is either binary or weighted, and $\mathbf{H} \in \mathbb{R}_+^{k\times n}$ is the community indicator matrix that is shared across all layers. Note that $\mathbf{H}, \mathbf{S}^{(i)}$ are nonnegative matrices. Furthermore, to overcome the unpredictability of unobserved entries, a binary weight matrix $\mathbf{W}$ is introduced to deal with missing values and is formed with respect to the following rules:

$$\forall i, \quad \mathbf{W}_{xy}^{(i)} = \begin{cases} 1 & \text{if } \mathbf{A}_{xy}^{(i)} \text{ is observed,} \\ 0 & \text{otherwise} \end{cases} \quad [5]$$

It is explicated in (6) that $\mathbf{H}$ in Equation 4 can be determined through solving an optimization problem defined in 6 using a trade-off parameter η_i to balance the equation:

$$\min \mathcal{L}(\mathbf{H}, \mathbf{S}^{(i)}) = \sum_{i=1}^{N} \|\mathbf{W}^{(i)} \circ (\mathbf{A}^{(i)} - \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T)\|_F^2 + \sum_{i=1}^{N} \eta_i \|\mathbf{S}^{(i)}\|_1 \quad [6]$$

Approaches are provided in (6) for optimizing with natural gradient and finding expressions for multiplicative updates for $\mathbf{H}$ and $\mathbf{S}^{(i)}$, which are 7 and 8.

$$\mathbf{S}^{(i)} \leftarrow \mathbf{S}^{(i)} \circ \sqrt{\frac{\mathbf{H}^T(\mathbf{W}^{(i)} \circ \mathbf{A}^{(i)})\mathbf{H}}{\mathbf{H}^T(\mathbf{W}^{(i)} \circ \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T)\mathbf{H} + \frac{1}{2}\eta_i}} \qquad [7]$$

$$\mathbf{H} \leftarrow \mathbf{H} \circ \left(\frac{\sum_{i=1}^{N}(\mathbf{W}^{(i)} \circ \mathbf{A}^{(i)})\mathbf{H}\mathbf{S}^{(i)}}{\sum_{i=1}^{N}(\mathbf{W}^{(i)} \circ \mathbf{H}\mathbf{S}^{(i)}\mathbf{H}^T)\mathbf{H}\mathbf{S}^{(i)}} \right)^{\frac{1}{4}} \qquad [8]$$

With this knowledge, we can further explore the clustering pattern of the AWN dataset while retaining more of its structural information by organizing observations of different days into different layers and freely setting the number of communities we wish to be detected.

Disease Spread Simulation. To further explore the community structure and develop novel centrality measures, disease spread experiments are conducted on the wildbird network. On the synthetic 100-day dataset, one or several initial "patient zero" nodes are selected to seed the infection. From the infected node, a random walk is performed to identify neighboring nodes to infect, with a probability proportional to the product of the weight of the connecting edge and a infection rate hyperparameter. Newly infected nodes are added to the set of infected nodes for the next iteration. The random walk process repeated from all infected nodes until the maximum simulation day is reached.

Infection Index. We define a take-off point of massive infection, where the disease spread exhibits faster than linear growth in the number of infected nodes over time. Formally, let N(t) be the number of infected nodes at time t. We identify the start of a massive infection, M, as the earliest time where:

$$\frac{N(t+1) - N(1)}{t-1} > 1$$

With each node as the patient zero node, we ran infection simulation for ten times and computed the average massive infection start time. To further quantify the significance of the patient zero node, we define an infection index metric calculated as:

$$1 - \frac{M - M_{min}}{M_{max} - M_{min}}$$

where M_min and M_max are the minimum and maximum values of M across all nodes. This metric is normalized between 0 and 1, with higher values indicating higher centrality of the node and earlier onset of massive infection that it leads to.

Basic Centrality Measures. The infection index showed a significant positive correlation with centrality measures across nodes. In different trials, most centrality measures exhibited Pearson and Spearman coefficients between 0.1 and 0.3 with the infection index we defined earlier. This demonstrates the relevance of our defined infection index. The more important a node is, the higher its centrality, and the more efficiently it can infect the entire network. It is worth noting that Katz Centrality showed considerable fluctuation in performance, and is therefore not considered a meaningful measure in this particular case.

Temporal Centralities. Several temporal centrality measures have been developed to address problems specifically in dynamic networks. One such measure is the Temporal Katz Centrality (7), a dynamic graph centrality measure designed to account for the temporal aspect of interactions in a network. It incorporates time-decay functions to emphasize the relevance of more recent interactions. It is defined as follows:

$$r_u(t) = \sum_{v} \sum_{z \in Z(v,u)} \phi(z,t) \qquad [9]$$

where $Z(v,u)$ denotes the set of temporal paths from node v to node u, and $\phi(z,t)$ is the weight of the temporal path z at time t.

The weight of a temporal path z is calculated as:

$$\phi(z,t) = \prod_{i=1}^{j} \varphi(t_{i+1} - t_i) \qquad [10]$$

The time decay function $\varphi(\tau)$ is defined as:

$$\varphi(\tau) = \beta \cdot \exp(-c\tau) \qquad [11]$$

Substituting the time decay function into the path weight formula, we get:

$$\phi(z,t) = \beta^{|z|} \cdot \exp\left(-c \sum_{i=1}^{j}(t_{i+1} - t_i)\right) \qquad [12]$$

Combining the above equations, the temporal Katz centrality becomes:

$$r_u(t) = \sum_{v} \sum_{z \in Z(v,u)} \beta^{|z|} \cdot \exp\left(-c \sum_{i=1}^{j}(t_{i+1} - t_i)\right) \qquad [13]$$

Another measure is the Temporal Betweenness Centrality (8), which calculates how often a node appears on the shortest paths between other nodes, considering the timing of interactions. The shortest path takes into account both physical distance and temporal distance. While betweenness centrality might not be the most suitable for our scenario, this particular formulation could actually be helpful.

Given a window W, the corresponding window graph is $G_W = (V, E_W)$. Let $\sigma_{SFP}(s,d)$ denote the number of SFPs from node s to node d, and $\sigma_{SFP}(s,d|v)$ denote the number of SFPs from s to d that pass through node v. The TBC of a node v is defined as:

$$TBC(v) = \sum_{s \neq v \neq d} \frac{\sigma_{SFP}(s,d|v)}{\sigma_{SFP}(s,d)}$$

Time-Weighted Degree Centrality and Development. Given the data format we are using, we have defined our own centrality measures in the hope that these would better capture a bird's importance in spreading disease. One such measure is the Time-Weighted Degree Centrality (TWDC). The TWDC C_i for a node i is defined as:

$$C_i = \sum_{j \in N_i} \sum_{(d,w) \in \text{weights}(i,j)} \frac{w}{d^{\alpha}}$$

where C_i is the Time-Weighted Degree Centrality of node i, N_i is the set of neighbors of node i, weights(i, j) is the set of weights associated with the edge between nodes i and j, w is the weight of the edge, d is the temporal distance associated with the weight, and α is a parameter that adjusts the influence of the temporal distance.

However, this only accounts for primary interactions. To better capture importance, we developed a recursive version of TWDC to account for secondary interactions from previous timestamps, naming it Time-Weighted Degree Centrality with Decay. The TWDC with Decay $C_i(t)$ for a node i at time t is defined as:

$$C_i(t) = \sum_{j \in N_i(t)} \frac{w_{ij}(t)}{t} + \gamma \sum_{j \in N_i(t-1)} \frac{w_{ij}(t-1)}{t-1} \cdot C_j(t-1)$$

where $C_i(t)$ is the Time-Weighted Degree Centrality with Decay of node i at time t, $N_i(t)$ is the set of neighbors of node i at time t, $w_{ij}(t)$ is the weight of the edge between nodes i and j at time t, and γ is a decay factor that adjusts the influence of the previous time step.

The cumulative Time-Weighted Degree Centrality with Decay over all time steps T is given by:

$$C_i = \sum_{t=1}^{T} C_i(t)$$

Test TWDC on Community. To further test the validity of TWDC, we combine it with the community structure of the dataset to run simulations. With community labels developed by above community detection algorithms, we evaluate the disease spread simulation under two cases of initial patient zero node selection: one selects the node with the highest TWDC in each of the communities as the patient zeros while the other one randomly choose the same number of patient zeros across the network. For each case, we ran 20 independent simulations and analyzed the progression of the disease spread. Our hypothesis is that if TWDC is a valid centrality measure, high-centrality nodes, by virtue of their stronger and more frequent interactions within communities, will lead to faster infection progression compared to randomly chosen patient zeros.

Three time points are set up to evaluate the progression of the disease spread. The first one is the start of the massive infection (M), and the other one is fully-infected date, the time when all of the 202 nodes are infected. Besides, since the cumulative curve of infections over time exhibits a sigmoid-like pattern, a sigmoid curve is fitted to the data. Using the bisection method, the halfway point, when the value reaches 101 and where the maximum infection speed occurs, is identified.

Results and Analysis

Community Detection. To get a baseline, we first tested several single-layer community detection algorithms as shown in Figure 1, summing up the weights observed during the entire 6-day window between each pair of nodes. We could tell that a partition of 6 communities generally gives the highest modularity in regardless of the choice of algorithm, all of which are above 0.3, the approximate cutoff for a valid clustering pattern as provided by Newman. This initial result is somewhat unexpected, because we naturally expected the number of communities in the best partition to align with the number of locations where the wildbird interactions are observed, which is 3.

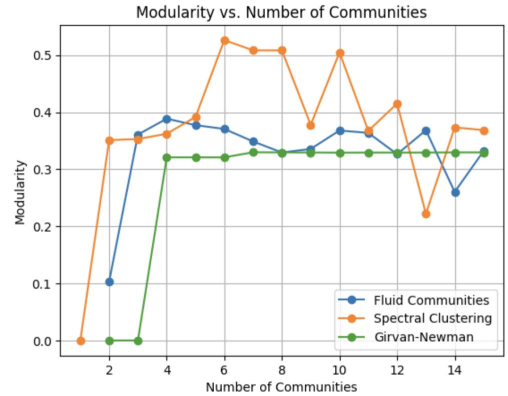

Fig. 1. Modularity of partition with different number of communities

To check the validity of this initial partition result, we compared it with the results given by the WSSNMTF multilayer algorithm introduced before, inputting the adjacency matrices of each day as separate layers. With a stop criterion of 0.01 for the iterative process of finding the community indicator matrix $\mathbf{H}$, we got the modularity results as shown in Table 1. This multilayer method gives lower modularity scores in general due to the fact that not all individuals are observed at each day and the binary weight matrix does not perfectly process this missing information on connectivity by simply setting unobserved weights to 0. However, the modularity results in Table 1 implies that a partition of 3 communities is relatively more meaningful, which is compatible with out initial expectation.

The disparity in the results given by the baseline single-layer methods and the multilayer detection algorithm shows an interesting connection, reflecting the different capability of embedding and retaining structural information of these algorithms, although the multilayer method is subject to unobserved weights that yield low modularity scores. Due to the considerable time complexity of this algorithm, we expect to perform a fine tuning of the parameters and improve the algorithm based on the actual dataset in future opportunities.

Number of Communities	Modularity of WSSNMTF
2	0.27245
3	0.24443
4	0.21426
5	0.18987
6	0.19715
7	0.21787
8	0.18372

Table 1. Modularity of WSSNMTF for different number of communities

Building on the observations from our baseline assessments and multilayer analysis, it becomes imperative to scrutinize how these detected community structures evolve under the stress of a spreading epidemic. The observed disconnect

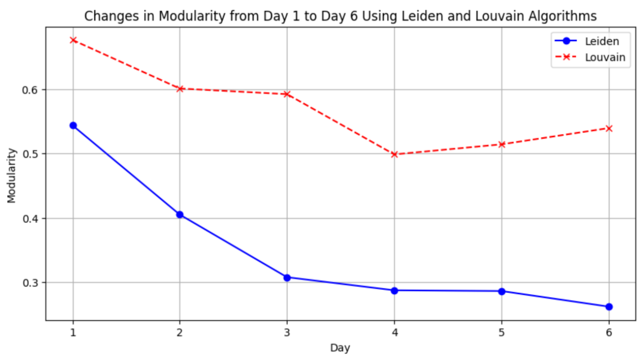

Fig. 2. Modularity of Division of the Network

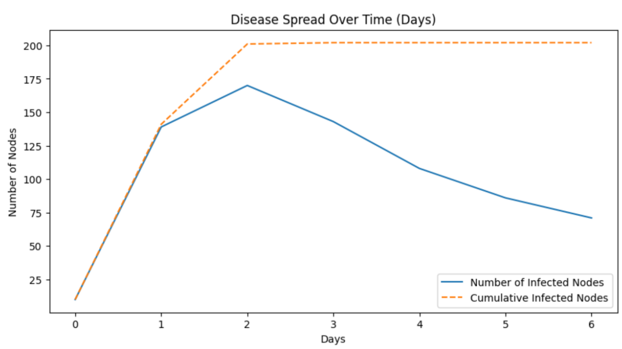

Fig. 3. Infected Bird Number of Each Day

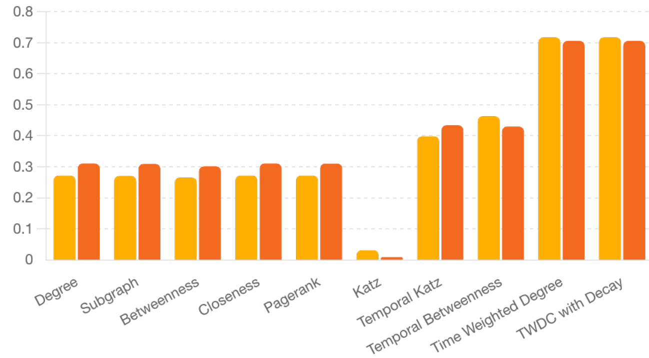

Fig. 4. Comparison of Centrality Measures' Correlation Coefficients with Infection Index

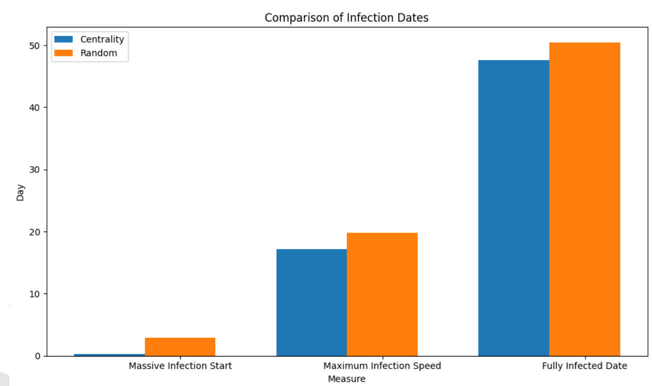

Fig. 5. Comparison of the three metrics of infection progression, the start of massive infection, maximum infection speed date, and fully infected date, in simulations with random patient zeros versus with high-centrality patient zeros.

between the expected and derived number of communities sets the stage for a deeper exploration into the dynamics of network transformation during critical events. In this context, Figure 2 tracks the changes in modularity, marked by a significant decrease, particularly with the Leiden algorithm, suggesting a substantial restructuring of the network's community bonds as the epidemic unfolds. Figure 3 tracks the disease spread, displaying a peak in the number of infected nodes. The most striking result is the observed steep decline in modularity during Days 1 and 2, paralleled by a significant increase in the number of infected birds. This suggests a correlation between community structure and disease spread, potentially due to a sudden shift in the structure of the network. Our results underscore the initial intense spread within communities, followed by a stabilization phase as the disease permeates through the broader network. The correlation between modularity decline and peak infection rates underscores the importance of examining the network structure during periods when major outbreaks occur.

Centrality. In general, the temporal centralities showed higher correlation coefficients with the infection index than the basic ones. This is because an interaction at a later time has a different significance than an interaction happening right now. Temporal dynamics capture the evolving nature of networks, reflecting how influence and information spread in real-time. Temporal centralities here allow us to account for these nuances. By incorporating the consideration of time, we can handle the weights more effectively. If we only map them into a static network, each edge might actually have multiple weight attributes, and simply adding them up could result in loss of information.

TWDC and TWDC with Decay work significantly better than other dynamic centralities. From our experiments, TWDC with Decay and TWDC consistently performed almost equally well. This is likely due to our use of a basic infection simulation model, the SI model, which excludes many other factors. We are also using a constant infection rate that is proportional to the weight of interaction. If we introduce more variability and factors into the model, we are likely to benefit from the additional degree of freedom provided by TWDC with Decay.

Centrality and Community. To test the new centrality measure based on community structure, the Louvain algorithm, the best performance community detection that we mentioned before, is used to identify 6 communities in our wildbird interaction network dataset. We run simulation under the high-TWDC patient zeros and the random patient zeros cases. Figure 5 presents a comparison of the three measures characterizing the infection progression between high-TWDC and random patient zero cases. The simulations initialized with high-TWDC patient zeros exhibited smaller values for all three metrics compared to the random case. This indicates a faster overall infection progression when the initial patient zeros corresponded to high-centrality nodes within their respective communities. The results align with our hypothesis and provide evidence supporting the validity of the TWDC measure.

Conclusion and Discussion

Our study on the dynamics of disease spread introduces novel centrality metrics and advancing advice on community detection algorithms. The development of Time-Weighted Degree Centrality with Decay successfully incorporates temporal dynamics into traditional centrality measures, enabling a more accurate representation of the importance and influence of individual nodes within the network. Through infection simulations, we have validated the effectiveness of our combined approach, which underscores the critical role of these temporal metrics and community structures in accurately predicting disease spread and designing effective wildlife epidemic control strategies. The findings highlight the dynamic interplay between modularity changes within communities and the timing and intensity of disease outbreaks, emphasizing how sudden shifts in network structure can precipitate and influence the spread of disease.

While our study provides valuable insights into disease spread dynamics in wild bird populations, it is important to acknowledge its limitations, primarily due to the constraints of the simulation models and the synthetic nature of the extended dataset. Our simulations assumed a constant infection rate and employed the simplest SI model, which might have oversimplified the complex nature of disease transmission in wildlife populations. The model does not account for potential recovery, immunity, or vaccination, which could significantly influence the spread dynamics. Future research could explore more sophisticated epidemiological models with varying infection rates. Expanding the developed methodologies to different types of networks and real-world datasets would help validate and refine our findings. Furthermore, integrating machine learning approaches could enable the identification of more nuanced patterns and relationships within and between communities during an epidemic.

Code Repository

Our works are saved at the drive.

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Author Contributions

Z.Z. and Y.W. designed research; all authors performed research; D.L., Y.C., Z.Z., Y.W. analyzed data; Z.Z. Y.C., D.L., and Y.W. contributed to methods and materials; Y.W. and S.S. wrote the paper results and conclusion; G.M. and S.S. wrote the abstract and some analysis.

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